

NERF PHOTOGRAMMETRY

NEURAL

RADIANCE

FIELDS

NERF

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WHAT IS NERF?

"Inverse Rendering" technology that uses Artificial Intelligence to transform 2D photos and videos into photorealistic, volumetric 3D experiences.

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NeRF does not create solid surfaces. He “sees” space as a volume of density (like fog). This allows you to represent glass, smoke and reflections, as calculates what is in and through the object, not just the shell.

To create the final image, the computer shoots “rays” of virtual light through this volume. The ray passes through the object, accumulating color and density along the way to form the final pixel on the screen.

Instead of storing geometry, NeRF stores a function mathematics. Receives coordinates (XYZ + Direction of Gaze) and the generative model returns the color and density at that exact point. It is a “Color Coordinate”.

CONCEPTS

FUNDAMENTALS

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Computer graphics transitioned from Explicit Geometry

(manual surface modeling) for Representation

Implicit (AI learning of light and volume).

This change allows you to capture the complexity of the world

real, such as reflections and transparencies, which traditional 3D does not
could replicate.

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FRAMEWORK

THEORETICAL

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Capturing Light (View

Dependency):

In classical theory, the color of an object is static (ink).

In NeRF, the color is Dynamic (View Dependent). The AI learn that the color of a point changes depending from where the observer is. (This is the theoretical basis to achieve photorealism in metals and glass.)

From "Explicit" to

"Implicit":

Before (Photogrammetry): 3D was based on Explicit Geometry (triangles and meshes). O computer needed to know exactly where the surface began and ended. Failed to smooth or transparent surfaces.

Now (NeRF): 3D becomes Representation Implicit. The object is not saved as a form, but rather as a "memory" in the network neural. The form "appears" only when we asked the AI.

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NeRF solves the inverse scientific problem: "How is that I discover the 3D shape of an object having just 2D photos, without using geometry?" The answer was replace geometry with density probability.

The "Inverse" Problem
rendering":

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NeRF went from being just an academic curiosity to become a powerful business tool. Yours unique ability to capture reality with fidelity photography opened doors in industries where the "synthetic look" of traditional 3D was unacceptable. Today, it is the technology of choice whenever realism, texture and complex lighting are more important than pure geometry.

AREAS OF APPLICATION

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Virtual Production:

Replacement of green screens in cinema/advertising with photorealistic NeRF scenarios.

Luxury E-commerce

Visualization of products where texture and shine are essential.

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Medicine and Science:

Volumetric data analysis (TACs, clouds) using
NeRF.

Real Estate and Tourism:

Virtual visits where mirrors and windows work
really.

Although the underlying mathematics (neural networks, functions Gaussians) is complex, the workflow for the end user has been drastically simplified. Today, it is not You need to be a data engineer to create a model volumetric, all you need is a smartphone and the correct technique. O process boils down to turning raw video into assets digital technologies through four fundamental steps.

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HOW IT WORKS:

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PIPELINE:

The Process:

Capture: Take a series of photographs (20 to 100+) around the object, ensuring that each photo overlaps the previous one.

Upload: Upload the original images (preferably in high resolution) directly to the cloud software.

Automatic Processing:

The software calculates the position of the cameras.

If it is NeRF: Train the neural network.

If Splatting: Generates and optimizes the point cloud.

Export:

Creation of an interactive web file (HTML/WebGL) or video.

Good Practices:

Overlap

Parallax

Sharpness

Lighting

Who Uses it?

How to do it?

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THE REAL ECOSYSTEM:

USES AND TOOLS

Rather than being a futuristic technology, NeRF is already integrated into strategies from big brands and accessible through tools simple.

McDonald's ("Hall of
Zodiacs"):

Google Maps

(Immersive View):

Used NeRF to create a

New Year's VR experience

Lunar, with complex lighting

that would be impossible in 3D

traditional mobile.

The ultimate example. Use Block-

NeRF to merge Satellite and Street

View, allowing you to fly over

cities like London or Lisbon

with realistic atmosphere.

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“USE CASES”

Meta (Avatars Codec):

Adidas & Rivian:

Creating hyper-avatars

realistic for the Metaverse,

capturing skin and expressions

real ones instead of dolls.

They use it for "Digital Twins". A

Adidas to show the texture

fuzzy sneakers; to Rivian

to show cars in forests

realistic where the light reflects on the

painting.

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“USE CASES”

For Editing

(Cleaning): SuperSplat
and PlayCanvas.

For Creators

(Simple): Luma AI and
Polycam (Web/iOS) and
Kiri Engine (Android)

Work like "Photoshop"
to clean up blemishes and edit
3D scenes in the browser.

Allows you to create scans with the
cell phone in minutes.

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TOOLS

For Professionals

(High-End): Instant

NGP (NVIDIA) and

Teleport by Varjo.

Focused on maximum quality

for VR and production

cinematographic.

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TOOLS

In addition to the novelty factor, the adoption of these technologies responds to critical challenges in e-commerce and digital communication. It's not just about aesthetics, but about a strategic tool that directly impacts metrics (KPIs) such as conversion rate, time to permanence on the site and the operational efficiency of the marketing.

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RELEVANCE TO

THE MARKETING

DIGITAL

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Reduction in Returns:

NeRF eliminates surprise. By capturing the texture and real imperfections (photorealism), the client knows exactly what you are buying.

Data: Fewer returns in e-commerce mean direct profit margin protection.

The Death of the Product Photo:

We are seeing the end of static photography product. Consumer expectations are shifting towards "Try before you buy" digitally.

Attention Economy and UX:

The modern consumer ignores static photos ("banner blindness"). An interactive object creates active engagement (touch, rotation), increasing the time spent in the website (Dwell Time) and SEO.

Concept: "Gamification without games".

COPE Strategy:

"Create Once, Publish Everywhere". A single NeRF scan generates: the 3D viewer for the website, the AR filter for Instagram and the VFX assets for the TV ad.

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NeRF

POTENTIAL IMPACT

(MARKET, SOCIETY,

EVOLUTION)

NeRF democratizes 3D, turning smartphones into studios professionals, but raises ethical questions about "Deepfakes of Locations" and privacy. Professionally, it requires a new fusion between engineering and art skills, creating essential hybrid profiles on the market.

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CONCLUSION AND

REFLECTION

NeRF = Brain/Compression/Atmosphere.

Gaussian Splatting = Speed/View

Web.

Photogrammetry = Physical Structure.

Success depends on Capture (blur-free video)

and using the right tool for the objective

right.

The "Sweet Spot" (The Future is Hybrid): We shouldn't

choose just one. Optimal flow uses Photogrammetry

for base, NeRF for intelligence and Splats for

visualization.

Current Limitations:

a.

Static

b.

Editing

Verdict: An indispensable presentation tool

of products, but still evolving to create worlds

complex interactives.

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THANK YOU

UNTIL NEXT TIME